

BrightSign NPU Platform - Executive Summary Slides

Slide 1: BrightSign NPU Platform

Next-Generation Edge AI for Digital Signage

Three-Tier Platform Architecture

Tier	Player	NPU	Capabilities
Premium	XT5	RK3588 (3-core, 6 TOPS)	3 models parallel - Full behavioral analytics
Mid-tier	XS156	RK3576 (2-core, 4 TOPS)	2 models parallel - Core engagement metrics
Entry	LS5/HS5	RK3568 (1-core, 1 TOPS)	1 model - Basic attention tracking

Built on proven BrightSign reliability:

- Fanless operation with 100,000-hour MTBF and 5-year warranty
 - 5.5W full AI workload (XT5) - 42-54% less power than competitors
 - C++ optimized implementation - 3.3x more efficient than Python alternatives
 - 3.6-4.5x better reliability than fan-cooled platforms
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Slide 2: BrightShopper - Flagship Application

Real-Time Retail Analytics at the Edge

What It Does:

- **People counting & tracking** - Movement patterns and traffic flow
- **Gaze detection & attention** - What content captures attention, for how long
- **Pose estimation (XT5)** - 17-keypoint skeleton tracking
- **Shopping behaviors** - Cart pushing, basket carrying, shelf interaction, product selection
- **Content attribution** - Link engagement to specific media assets (BSN.cloud)

How It's Different:

- **Edge-first, privacy-first** - All processing on-device, zero PII transmitted
 - **Real-time performance** - Millisecond latency vs. seconds for cloud
 - **Dual-purpose hardware** - Digital signage + analytics in one device
 - **No cloud costs** - No bandwidth, no recurring fees
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Slide 3: Competitive Advantages

Why BrightSign NPU Platform Wins

vs. Cloud-Based Analytics

- **Zero latency:** Milliseconds vs. seconds for round-trip
- **Privacy compliance:** No video leaves device, no PII collected
- **100% uptime:** Works without internet connectivity

vs. Python Edge AI Implementations

- **3.3x lower power per NPU core:** C++ vs. Python efficiency
- **Memory bandwidth:** Python saturates NPU at 3 models; C++ enables true parallelism
- **Production reliability:** Compiled binary vs. interpreter dependencies

vs. External NPU Accelerators (Hailo, Coral, etc.)

- **1.8x lower total power:** No USB/PCIe overhead, native integration
- **Simpler design:** PoE-powered single-box solution

vs. Dedicated Retail Analytics Cameras

- **Lower TCO:** No separate camera infrastructure
- **Content attribution:** Automatic linkage of viewer data to displayed media

Key Metrics (XT5):

- **0.42 TOPS/Watt** - 1.5-2.3x better computational efficiency
- **0.84 R-TOPS/Watt** - 7-8x better lifetime computational value
- **\$62K-\$100K savings** over 5 years per 1,000 units deployed